Â§11.4.108 Four-Layer Verification â€” Session Commits 457cca4..fab6707

Revision: 1 **Last modified:** 2026-08-18T20:50:00Z

Scope

Verifies the ~28 commits landed in this session (`457cca4` -> `fab6707`, 81 files changed) against the running boba stack, per Â§11.4.108 (SOURCE -> ARTIFACT -> RUNTIME-ON-CLEAN-TARGET -> USER-VISIBLE).

Phase 1 â€” Runtime-impact categorization

`git diff --name-only 457cca4..fab6707` enumerated. Every changed file classified:

Category	Files	Runtime impact
C â€” rebuild required	`qBitTorrent-go/internal/jackettapi/health.go`, `health_test.go`	YES â€” boba-jackett is `build: context: ./qBitTorrent-go, dockerfile: Dockerfile.jackett` in `docker-compose.yml` (not a bind-mount)
B â€” host-level, non-container	`scripts/install-resource-pressure-timer.sh` + `scripts/systemd/user/boba-resource-pressure-check.{service,timer}`	NO â€” systemd `--user` timer, entirely outside the docker-compose stack
A â€” no runtime impact	Everything else: `docs/**`, `challenges/**`, `scripts/commit-push-all.sh`, `scripts/install-dev-tools.sh`, `scripts/capture-workable-items-db-delta.sh`, `scripts/pre_build_verification.sh`, `scripts/host-power-management/check-no-suspend-calls.sh`, `constitution` (submodule pointer)	NO â€” dev tooling / challenge scripts / docs / governance, none container-mounted or container-built

Verdict: exactly one runtime-affecting source pair (BOB-112â€™s TTL-cache fix in `health.go`). No `docker-compose.yml`, no `Dockerfile.jackett`, no `download-proxy/src/`, no `plugins/`, no `config/` changes this session.

Phase 2 â€” Rebuild + reflash

Finding before acting: the *already-running* `boba-jackett` containerâ€™s image (`962f4452c725...`) was built at `2026-08-18T19:06:17Z` â€” **after** `health.goâ€™s` last touching commit `91b52db` (`19:02:50Z`) â€” because a prior subagent in this same session (commit `e2a2e3e`, evidence at `docs/qa/BOB-112/`) had already rebuilt + RED/GREEN-mutation-verified this exact fix with live `wrk` load tests (97.1%->0.0% timeout rate, full methodology + numbers in `docs/qa/BOB-112/summary.md`). Confirmed no commit after `91b52db` touched `health.go` (`git log 91b52db..fab6707 -- health.go` -> empty).

This taskâ€™s job was therefore to **redeploy on a genuinely clean target** (Â§11.4.108 clause 4 â€” eliminate the stale-deployment/shadow layer by construction) and confirm the fix holds, rather than repeat the prior subagentâ€™s RED/GREEN mutation study.

Ran `./start.sh --recreate` (the CLAUDE.md-sanctioned Level-3 restart for â€œdocker-compose.yml, start-proxy.sh, env var, base imageâ€ changes â€” the correct level given a rebuilt image needed a fresh container instance). Full transcript: `start_sh_recreate.log`.

- Tracker-cookie loader ran first (per operator mandate 2026-08-15): all 5 trackers (rutracker/nnmclub/rutor/kinozal/iptorrents) reported `UNCHANGED (idempotent)` â€” no drift, no re-extraction needed.
- `boba-ctl` (Go orchestrator) executed `podman-compose ... down` then `... up -d` â€” exit 0 both steps.
- Only bobaâ€™s own 4 services (`qbittorrent`, `jackett`, `qbittorrent-proxy`, `boba-jackett`) were touched. Confirmed via before/after `podman ps -a` (`container_status.log`) that the 15 unrelated containers belonging to other projects on this host (`proxy-*`, `lava-*`, `helix-*`, `deploy_caddy_1`, `wonderful_dijkstra`) were **not restarted, not recreated, not touched** â€” their `Up 2 hours` / `Up 12 hours` uptimes are unchanged across the before/after snapshots.
- All 4 boba containers reached `healthy` within ~1 minute (`qbittorrent/jackett`: healthy at 52s; `boba-jackett/qbittorrent-proxy`: healthy at 46s).
- Post-recreate cookie lengths (byte counts only, Â§11.4.10 â€” no values logged) are **byte-identical** to the pre-recreate baseline for all 5 trackers (`cookie_state_post_recreate.log`).

Phase 3 â€” Runtime verification

All 5 endpoints reachable post-recreate (`endpoint_health.log`, verbatim):

Endpoint	Result
`GET :7186/api/v2/app/version` (qBittorrent via proxy)	`200`, body `v5.2.3`
`POST :7186/api/v2/auth/login` (admin/admin)	`204` (qBittorrentâ€™s own success response)
`GET :7187/health` (merge service)	`200`, `{"status":"healthy","service":"merge-search","version":"1.0`
`GET :7189/healthz` (boba-jackett, NEW TTL cache)	`200`, `{"status":"ok","db_ok":true,"jackett_ok":true,"version":"0.1.0"`
`GET :9117/api/v2.0/server/config` (Jackett upstream)	`302` (redirect to admin auth â€” reachable, expected for an unauthenticate

BOB-112 fix â€” live verification on the fresh container

Artifact layer (`artifact_layer_check.txt`): the BOB-112 fixâ€™s unique `log.Printf` format string (added in `91b52db`, absent in every prior version of `health.go`) is present in the **freshly-recreated** containerâ€™s binary:

```
boba-jackett: /healthz Jackett cache refresh #%d (hits=%d misses=%d ratio=
```

Image digest `962f4452c725...` â€” identical to the digest the prior subagentâ€™s RED/GREEN mutation study (`docs/qa/BOB-112/summary.md`) recorded as the confirmed post-fix, post-revert-verification build. No drift.

Runtime layer (`wrk_healthz_evidence.log`): brief `wrk -t2 -c50 -d5s --timeout 3s --latency http://localhost:7189/healthz` against the container that had been up for well under a minute (genuinely cold cache at test start):

```
270085 requests in 5.00s, 48.17MB read
Requests/sec:  54008.16
Latency: avg 1.08ms, p50 844us, p90 2.31ms, p99 4.42ms
```

0 timeouts / 0 socket errors (wrk prints no â€œSocket errorsâ€¦ line when all three counters â€” connect/read/write/timeout â€” are zero) â€” well under the taskâ€™s <5% threshold, and consistent with the prior subagentâ€™s full 30sx100c study (0.0% timeouts, 27k req/s).

Container logs (`podman logs boba-jackett`) show exactly 2 upstream `Jackett.GetCatalog()` calls across the whole test window â€” one at boot (`cache refresh #1`, cold cache) and one ~31s later once the 30s TTL first expired â€” confirming the cache, not luck, is what collapsed 270k `/healthz` hits into 2 real upstream calls.

Honest gaps / not covered by this task

- **No image rebuild was forced.** `./start.sh --recreate` is `compose down && compose up -d` â€” it does not pass `--build`, so it reused the existing `boba_boba-jackett:latest` image rather than compiling a fresh one from current `HEAD` (`fab6707`). This is not a gap for BOB-112 specifically (proven byte-identical via the format-string + digest match above, and `git log 91b52db..fab6707 -- health.go` is empty so there is nothing newer to rebuild), but it means this task did not independently *reproduce* a build from source â€” it verified the artifact already known to be correct survives a clean container-level redeploy. A from-scratch `podman-compose build boba-jackett` was not run because it would have produced a byte-identical image to the one already running (no source delta since the prior subagentâ€™s own build), and CLAUDE.mdâ€™s Critical Constraints table has no sanctioned `start.sh` subcommand for â€œforce-rebuild a containerâ€™s Go source without a compose/Dockerfile/env changeâ€ (only `--reload-python`, `--reload-plugins`, and `--recreate` exist) â€” using a raw `podman-compose build` would have stepped outside the orchestrator-only mandate for a rebuild this taskâ€™s own evidence shows was unnecessary.
- **Â§11.4.185 manual QA (human confirmation) not performed** â€” out of scope for this automated verification subagent; this report is the automated Â§11.4.108 evidence layer only, not the terminal human sufficiency gate.
- **`qbittorrent-proxy` (Python/FastAPI merge service) source unchanged this session** â€” verified reachable (`:7187/health` 200) but not independently re-tested beyond the health endpoint, since no commit touched `download-proxy/src/`.
- The host-level `boba-resource-pressure-check.timer` (BOB-076/task-77) was spot-checked (`systemctl --user list-timers`, active, fired ~5 min prior) for completeness but is **not part of the boba container stack** and its own evidence trail already lives at `docs/qa/task-77/`.

Verdict

PASS â€” the one runtime-affecting change this session (BOB-112 TTL cache) is confirmed live on a genuinely fresh container instantiation via both

artifact-layer (binary string + image digest match) and runtime-layer (0%-timeout `wrk` run + exact cache-refresh-count log evidence) proof. All 5 boba-stack endpoints reachable post-recreate. Zero unrelated host containers disturbed. All 5 tracker cookies preserved byte-length-identical across the recreate.